Swapnil Pandey's Divisibility Rule for 3

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Abstract:

This document introduces a new and original divisibility test for 3 discovered by Swapnil Pandey. Unlike the conventional method which checks the sum of digits, this method leverages a transformation involving the unit digit and the rest of the number, providing an elegant and alternative approach to check divisibility by 3. The rule is mathematically validated and shown to work recursively for numbers of any length.

The Rule:

Let a number be written as $(N = 10Q + z)$, where:

- (z) is the **last digit** of the number.
- (Q) is the **rest of the number** (all digits except the last).

Then: (N Q + 4z)

Step-by-Step Method:

1. Take the **last digit** of the number (denote it as (z)).
2. Multiply it by **4**.
3. Add the result to the **rest of the digits** (denoted (Q)).
4. Check if the resulting number is divisible by 3.
5. **If not**, apply the rule again recursively until a small number is obtained.
6. If the final result is divisible by 3, so is the original number.

Example 1: 123

- Last digit $(z = 3)$
- Remaining digits $(Q = 12)$
- $(Q + 4z = 12 + 12 = 24)$
- 24 is divisible by 3 () 123 is divisible by 3.

2: 98765436 (8-digit)

Q = 9876543 → (Q + 4z = 9876567)

- z = 7, Q = 987656 → (987656 + 28 = 987684)
- Repeat: 98784 → 9894 → 1005 → 1 + 0 + 0 + 5 = 6
- 6 is divisible by 3 → So is 98765436

Conclusion:

This divisibility test presents a unique alternative to traditional digit-sum checks. It is elegant, recursive, and particularly useful for mental math and competitive exams. Swapnil Pandey's method, now titled **Pandey's Rule**, is mathematically sound and universally applicable.

Status:

- Verified ✅
- Suitable for mathematical publication or inclusion in competitive math training.
- Potential candidate for recognition in innovation and number theory education.

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